Software Requirements Specification (SRS)

Project: Collaborative Study Material & Quiz Platform (CSQP)

Version: 1.0

Authors: John S Thomas (PES2UG24CS207), Kritika Mahapatra (PES2UG24CS239), Krishna Manoj (PES2UG24CS236), Harshith R Reddy (PES2UG24CS188)

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Revision History

Version	Date	Author	Change Summary	Approval
1.0	05-09-2026	Team 4	Final SRS submitted for evaluation	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

Team Member Contribution

No.	Team Member	Assigned Work
1	Harshith R Reddy (PES2UG24CS188)	Introduction; External Interface Requirements
2	Kritika Mahapatra (PES2UG24CS239)	Overall Description; UML Use-Case Diagrams
3	Krishna Manoj (PES2UG24CS236)	System Features: Functional Requirements; Quality Attributes & Acceptance Tests
4	John S Thomas (PES2UG24CS207)	Non-Functional Requirements + Security; Requirements Traceability Matrix (RTM)

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Collaborative Study Material & Quiz Platform (CSQP), a web application that lets students share study material, build and attempt quizzes from a shared question bank, and track performance over time. It defines functional and non-functional requirements, interfaces, and verification criteria for the development and evaluation of the project.

1.2 Scope

Covers student and instructor interactions with the web platform for material sharing, question-bank management, quiz creation and attempts, and performance analytics. Includes the MERN-stack web application (React frontend, Node/Express backend, MongoDB, with selected services in Python for analytics) and its REST APIs. Excludes integration with external Learning Management Systems (LMS) and any institutional student-information system beyond basic account fields.

1.3 Audience

Developers (Team 4), Course Instructor/Evaluator, QA reviewers, and end users (students and instructors) piloting the platform.

1.4 Definitions

List of acronyms: MERN (MongoDB, Express, React, Node.js), UI (User Interface), API (Application Programming Interface), JWT (JSON Web Token), MCQ (Multiple Choice Question), CSV (Comma-Separated Values), RBAC (Role-Based Access Control), TLS (Transport Layer Security), WCAG (Web Content Accessibility Guidelines).

2. Overall Description

2.1 Product Perspective

CSQP is a new, standalone web application built on the MERN stack. It consists of a React single-page frontend, a Node/Express REST API backend, a MongoDB data store, and a Python-based analytics component that computes performance statistics. It is not a replacement for an institutional LMS; it is a supplementary collaborative study and self-assessment tool.

2.2 Major Product Functions

- User registration, login, and role-based access (Student / Instructor / Admin)
- Upload, search, rate, and version study material
- Build and maintain a shared, tagged question bank
- Create and configure quizzes from the question bank
- Attempt quizzes with timing, randomization, and anti-cheating controls
- Automatic and manual grading of quiz attempts
- Individual and class-level performance analytics and reporting
- Optional leaderboard and peer question contribution with approval workflow

2.3 User Roles and Characteristics

- Student: uploads/consumes material, attempts quizzes, contributes questions, views personal analytics.

- Instructor: manages the question bank, approves student-submitted questions, creates quizzes, views class-level analytics.
- Admin: manages user accounts and roles, monitors system health and audit logs.

2.4 Operating Environment

Browser-based client (Chrome, Firefox, Edge, Safari - latest two versions), responsive for desktop and mobile. Backend hosted on a cloud VM/container (Node.js runtime) with a managed MongoDB instance. Standard broadband/campus network connectivity assumed; no specialised hardware required.

2.5 Constraints

Must be implementable within a single academic semester by a 4-member student team using Python/JavaScript (MERN stack). Must comply with institutional data-handling expectations for student data. Free/low-cost hosting and storage tiers are assumed for the project deployment.

3. External Interface Requirements

3.1 User Interfaces

Primary UI: responsive web interface with distinct Student, Instructor, and Admin views. Accessibility: keyboard-navigable forms and sufficient colour contrast for core flows (login, quiz attempt, dashboard).

3.2 Hardware Interfaces

None beyond standard client devices (desktop/laptop/mobile/tablet) with a modern web browser and internet connectivity.

3.3 Software Interfaces

- REST API (Node/Express) over HTTPS/JSON between the React frontend and backend.
- MongoDB for persistent storage of users, materials, question bank, quizzes, and attempts.
- Email/OTP service (e.g., SMTP provider) for account verification and password reset.
- Object/file storage (e.g., local disk or S3-compatible service) for uploaded materials.

3.4 Communications

TLS 1.2+ enforced for all client-server traffic. API requests use short-lived JWT access tokens with refresh-token rotation (see Section 5.1).

4. System Features (Detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow CSQP-F-###.

4.1 Authentication & Account Management

Description: Register, authenticate, and manage access for Students, Instructors, and Admins.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSQP-F-001	The system shall allow a user to register using a college email address and verify the account via an emailed OTP/link before first login.	Functional	High	Students / Course Admin	AC-CSQP-F-001: Unverified accounts cannot log in; verification link expires in 24h. Test: TC-Auth-01	Requires Email/OTP service
CSQP-F-002	The system shall authenticate users via email/username and password and issue a signed JWT session token on success.	Functional	High	All Users	AC-CSQP-F-002: Valid credentials return a JWT; invalid credentials are rejected with a generic error. Test: TC-Auth-02	Depends on Auth service
CSQP-F-003	The system shall enforce role-based access control for three roles: Student, Instructor, and Admin.	Functional	High	Security / Course Admin	AC-CSQP-F-003: Endpoints reject requests where the caller's role lacks permission (HTTP 403). Test: TC-Auth-03	Applies to all API routes
CSQP-F-004	The system shall allow a user to reset a forgotten password via a time-limited, single-use emailed reset link.	Functional	Medium	Students	AC-CSQP-F-004: Reset link expires after 30 minutes or first use. Test: TC-Auth-04	Requires Email service

4.2 Study Material Sharing

Description: Upload, discover, and review peer-shared study material.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSQP-F-010	The system shall allow a Student to upload study material (PDF, DOCX, PPTX, images up to 25MB) tagged with subject and topic.	Functional	High	Students	AC-CSQP-F-010: Upload succeeds for supported types/size; rejected otherwise with a clear message. Test: TC-Mat-01	Storage service (e.g., S3/local)
CSQP-F-011	The system shall allow users to search and filter materials by subject, topic tag, uploader, and average rating.	Functional	High	Students	AC-CSQP-F-011: Filtered results match all applied criteria within 2s. Test: TC-Mat-02	Depends on search index
CSQP-F-012	The system shall allow Students to rate (1-5 stars) and comment on shared	Functional	Medium	Students	AC-CSQP-F-012: Rating updates the material's average	

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	materials.				score in real time. Test: TC-Mat-03	
CSQP-F-013	The system shall retain prior versions of a material when a newer version is uploaded, preserving version history.	Functional	Medium	Students / Instructors	AC-CSQP-F-013: Previous version remains downloadable after a new upload. Test: TC-Mat-04	Storage versioning

4.3 Quiz Creation & Question Bank

Description: Build and curate a shared question bank and assemble quizzes from it.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSQP-F-020	The system shall allow Instructors and Students (with approval) to add questions to a shared question bank, tagged by subject, topic, and difficulty.	Functional	High	Instructors / Students	AC-CSQP-F-020: New question is stored with required tags; appears in 'Pending Approval' if submitted by a Student. Test: TC-QB-01	Approval workflow (CSQP-F-023)
CSQP-F-021	The system shall support MCQ, True/False, and Short-Answer question types in the question bank.	Functional	High	Instructors	AC-CSQP-F-021: All three types can be created, edited, and rendered correctly in a quiz. Test: TC-QB-02	
CSQP-F-022	The system shall allow bulk import/export of questions via CSV/Excel with validation of required fields.	Functional	Medium	Instructors	AC-CSQP-F-022: Malformed rows are rejected with row-level error messages; valid rows are imported. Test: TC-QB-03	
CSQP-F-023	The system shall require Instructor approval before a Student-submitted question becomes usable in a quiz.	Functional	High	Instructors	AC-CSQP-F-023: Unapproved questions are excluded from quiz generation. Test: TC-QB-04	Depends on CSQP-F-020
CSQP-F-024	The system shall allow an Instructor to create a quiz by selecting or randomly sampling questions from the bank, with a configurable time limit and number of attempts.	Functional	High	Instructors	AC-CSQP-F-024: Generated quiz honours the selected/sampled question count, time limit, and attempt cap. Test: TC-QB-05	

4.4 Quiz Attempt & Grading

Description: Timed quiz attempts with automatic grading and anti-cheating controls.

Req ID	Requirement (shall...)	Type	Priority	Source/	Acceptance criteria /	Comments /

				Stakeholder	Test case ref	Dependencies
CSQP-F-030	The system shall present a quiz with a visible countdown timer and auto-submit answers when time expires.	Functional	High	Students	AC-CSQP-F-030: Attempt is auto-submitted at 00:00 with all answered questions saved. Test: TC-Att-01	
CSQP-F-031	The system shall automatically grade objective questions (MCQ, True/False) immediately upon submission and flag short-answer questions for manual review.	Functional	High	Students / Instructors	AC-CSQP-F-031: Score for objective items appears within 2s of submission; subjective items show 'Pending Review'. Test: TC-Att-02	
CSQP-F-032	The system shall randomize question order and option order per attempt and disable copy-paste on the quiz screen to reduce academic dishonesty.	Functional	High	Security / Instructors	AC-CSQP-F-032: Two students receive different question/option ordering for the same quiz. Test: TC-Att-03	Ties to security objective SEC-OBJ-02
CSQP-F-033	The system shall restrict a Student to a single active attempt session per quiz at a time, rejecting concurrent attempts from another device.	Functional	Medium	Students / Instructors	AC-CSQP-F-033: A second login attempting the same quiz is blocked while a session is active. Test: TC-Att-04	

4.5 Performance Analysis & Reporting

Description: Individual and class-level analytics derived from quiz attempts.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSQP-F-040	The system shall provide each Student a performance dashboard showing quiz history, score trends over time, and topic-wise strengths/weaknesses.	Functional	High	Students	AC-CSQP-F-040: Dashboard reflects all completed attempts within 5s of quiz submission. Test: TC-Perf-01	
CSQP-F-041	The system shall provide Instructors class-level analytics including average scores, most-missed questions, and participation rate per quiz.	Functional	High	Instructors	AC-CSQP-F-041: Analytics recompute after each new submission and match manually verified totals. Test: TC-Perf-02	
CSQP-F-042	The system shall provide an opt-in leaderboard ranking Students by score per quiz or subject.	Functional	Low	Students	AC-CSQP-F-042: Students who opt out are excluded from the visible leaderboard. Test: TC-Perf-03	
CSQP-F-043	The system shall allow Students and Instructors to export performance reports as PDF or CSV.	Functional	Medium	Students / Instructors	AC-CSQP-F-043: Exported file matches on-screen data at time of export. Test: TC-	

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
					Perf-04	

5. Non-Functional Requirements (Detailed)

NFRs below are measurable and tied to test plans. IDs follow CSQP-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
CSQP-NF-001	Page loads and quiz-submission responses shall complete within 2 seconds for 90% of requests under normal load (up to 500 concurrent users).	Performance	High	Load test shows p90 <= 2s. Test: TC-Perf-Load-01
CSQP-NF-002	The platform shall provide 99.5% availability monthly, excluding scheduled maintenance windows announced 24h in advance.	Reliability	High	Uptime monitoring report >=99.5%/month. Test: Ops dashboard review
CSQP-NF-003	All personal data and quiz content shall be protected according to the security requirements in Section 5.1. Passwords shall never be stored or logged in plaintext.	Security/Compliance	High	Security checklist pass. Test: TC-Sec-01..05
CSQP-NF-004	The web UI shall be responsive on desktop and mobile viewports and conform to WCAG 2.1 AA for core flows (login, attempt quiz, view dashboard).	Usability/Accessibility	Medium	Accessibility audit pass on core flows. Test: TC-UX-01
CSQP-NF-005	The backend shall be horizontally scalable to support at least 5,000 registered users and 500 concurrent quiz attempts without architectural change.	Scalability/ Maintainability	Medium	Load test at target scale passes NF-001 thresholds. Test: TC-Scale-01

5.1 Security

5.1.1 Security Objectives

- SEC-OBJ-01: Protect student personal data and academic records from unauthorized access, disclosure, or tampering.
- SEC-OBJ-02: Preserve the confidentiality and integrity of quiz content and results to prevent leakage and academic dishonesty.
- SEC-OBJ-03: Ensure the platform remains available and responsive during scheduled quiz windows.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
CSQP-SR-001	All client-server communication shall use TLS 1.2 or higher.	Security	High	AC-SR-001: Plaintext HTTP requests are rejected/redirected. Test: TC-Sec-01
CSQP-SR-002	User passwords shall be hashed using bcrypt (or Argon2) with a per-user salt; passwords shall never be stored or logged in plaintext.	Security	High	AC-SR-002: Database inspection shows only hashes; logs contain no raw passwords. Test: TC-Sec-02
CSQP-SR-003	Access (JWT) tokens shall expire within 1 hour and be renewed only via a rotating refresh token.	Security	High	AC-SR-003: Expired access tokens are rejected;

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
				refresh reissues a new token pair. Test: TC-Sec-03
CSQP-SR-004	Role-based access control shall be enforced at the API layer for every endpoint that returns or modifies user, material, or quiz data.	Security	High	AC-SR-004: Automated scan confirms no endpoint is reachable without an authorized role. Test: TC-Sec-04
CSQP-SR-005	The system shall lock an account for 15 minutes after 5 consecutive failed login attempts and record an audit log entry.	Security	Medium	AC-SR-005: 6th consecutive failure is rejected even with correct credentials; audit log shows the event. Test: TC-Sec-05

6. Quality Attributes & Acceptance Tests

Acceptance criteria: all High-priority functional requirements shall be implemented and verified during system testing. No critical NFR failures shall remain before final acceptance. The RTM in Section 8 tracks implementation/test status for every requirement (see status legend in Section 8).

Acceptance test suites: Authentication, Material Sharing, Question Bank, Quiz Attempt & Grading, Performance Analytics, Security, and Accessibility tests.

Quality Attribute	**Requirement**
Performance	90% of requests complete within 2 seconds
Reliability	99.5% monthly availability
Security	Passwords/PII protected per Section 5.1; no plaintext passwords
Usability	WCAG 2.1 AA for core flows
Scalability	500 concurrent quiz attempts

7. UML Use-Case Diagrams

Two use-case diagrams are provided below: one for the Student role, and one covering the Instructor and Admin roles.

Figure 7.1(a): Use-Case Diagram - Student

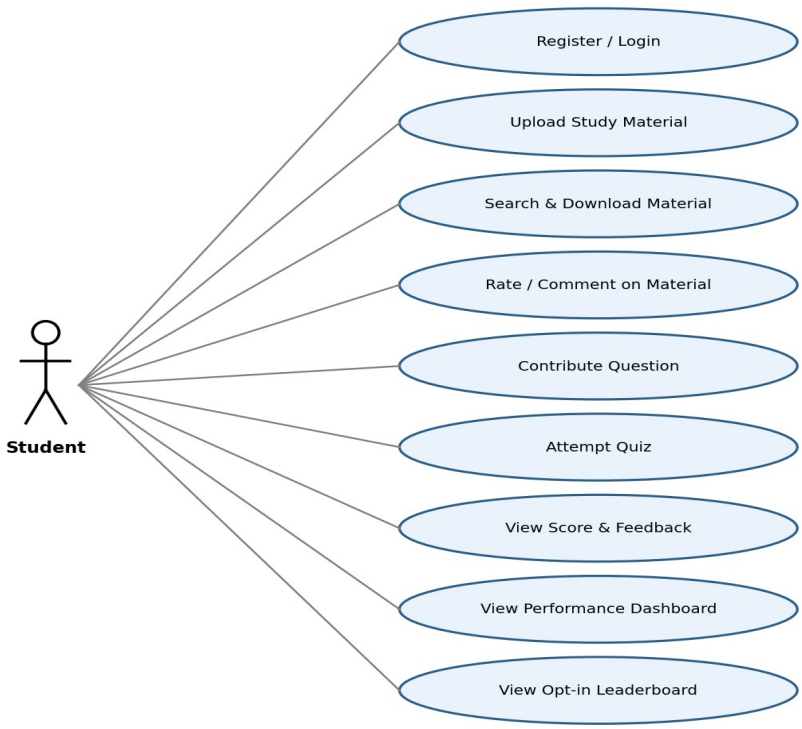


Figure 7.1(a): Use-Case Diagram - Student

Figure 7.1(b): Use-Case Diagram - Instructor / Admin

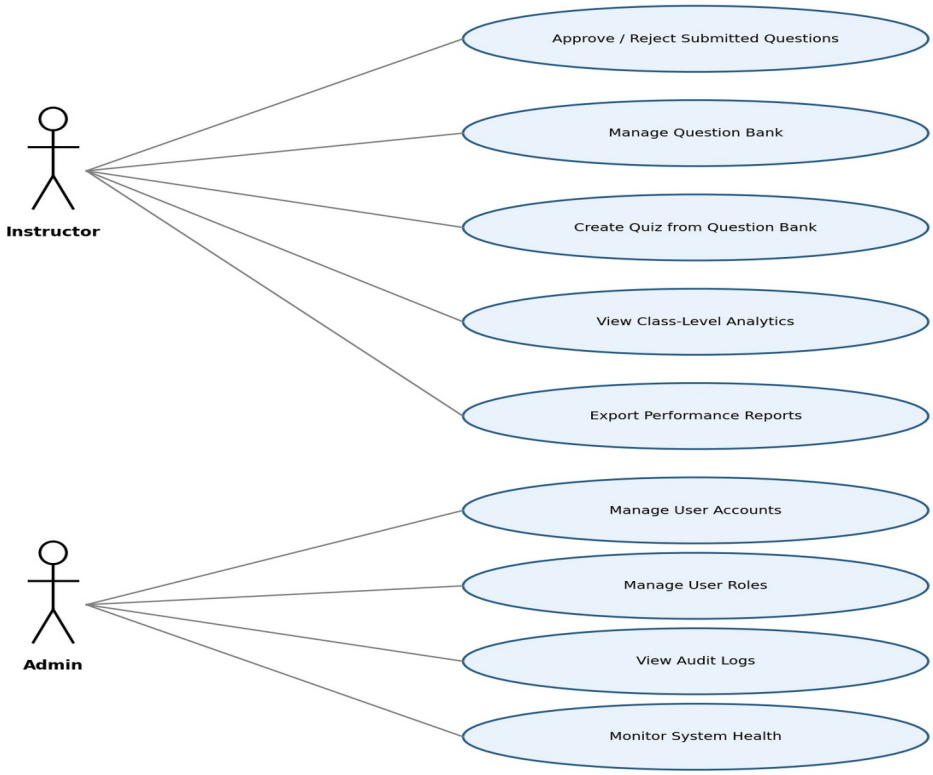


Figure 7.1(b): Use-Case Diagram - Instructor / Admin

8. Requirements Traceability Matrix (RTM)

Status legend: N = Not started, P = In progress, A = Accepted (implemented and verified). All items are currently N as this SRS is submitted prior to implementation and system testing.

Req ID	Requirement (short)	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
CSQP-F-001	College email registration & verification	4.1 / DS-Auth-01	AuthModule	TC-Auth-01	N	Email/OTP service
CSQP-F-002	Login / JWT auth	4.1 / DS-Auth-01	AuthModule	TC-Auth-02	N	Depends on Auth service
CSQP-F-003	Role-based access control	4.1 / DS-Auth-01	AuthModule	TC-Auth-03	N	Applies to all API routes
CSQP-F-004	Password reset	4.1 / DS-Auth-01	AuthModule	TC-Auth-04	N	Email service
CSQP-F-010	Upload material	4.2 / DS-Material-01	MaterialModule	TC-Mat-01	N	Storage service
CSQP-F-011	Search & filter materials	4.2 / DS-Material-01	MaterialModule	TC-Mat-02	N	Search index
CSQP-F-012	Rate & comment on materials	4.2 / DS-Material-01	MaterialModule	TC-Mat-03	N	—
CSQP-F-013	Material version history	4.2 / DS-Material-01	MaterialModule	TC-Mat-04	N	Storage versioning
CSQP-F-020	Add questions to question bank	4.3 / DS-Quiz-01	QuestionBankModule	TC-QB-01	N	Approval workflow
CSQP-F-021	Support multiple question types	4.3 / DS-Quiz-01	QuestionBankModule	TC-QB-02	N	MCQ, True/False, Short Answer
CSQP-F-022	Bulk import/export questions	4.3 / DS-Quiz-01	QuestionBankModule	TC-QB-03	N	CSV/Excel validation
CSQP-F-023	Approve student questions	4.3 / DS-Quiz-01	QuestionBankModule	TC-QB-04	N	Depends on F-020
CSQP-F-024	Generate quiz from question bank	4.3 / DS-Quiz-01	QuizModule	TC-QB-05	N	Configurable time/attempt limit
CSQP-F-030	Timed quiz & auto-submit	4.4 / DS-Quiz-01	QuizModule	TC-Att-01	N	—
CSQP-F-031	Auto-grade objective questions	4.4 / DS-Grading-01	GradingModule	TC-Att-02	N	Short answers require manual review
CSQP-F-032	Randomize questions/options	4.4 / DS-Quiz-01	QuizModule	TC-Att-03	N	Security objective SEC-OBJ-02
CSQP-F-033	Single active quiz attempt	4.4 / DS-Quiz-01	QuizModule	TC-Att-04	N	—
CSQP-F-040	Student performance dashboard	4.5 / DS-Analytics-01	AnalyticsModule	TC-Perf-01	N	—
CSQP-F-041	Class-level analytics	4.5 / DS-Analytics-01	AnalyticsModule	TC-Perf-02	N	Instructor access
CSQP-F-042	Student leaderboard	4.5 / DS-Analytics-01	AnalyticsModule	TC-Perf-03	N	Opt-in

Req ID	Requirement (short)	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
CSQP-F-043	Export performance reports	4.5 / DS-Analytics-01	AnalyticsModule	TC-Perf-04	N	PDF/CSV
CSQP-NF-001	Response time target	5 / DS-Perf-01	Web UI / API	TC-Perf-Load-01	N	p90 ≤ 2s
CSQP-NF-002	99.5% monthly availability	5 / DS-Perf-01	Infrastructure / Monitoring	Ops dashboard review	N	Scheduled maintenance excluded
CSQP-NF-003	Protect personal data & quiz content	5 / DS-Sec-01	SecurityModule	TC-Sec-01..05	N	See Security Requirements 5.1.2
CSQP-NF-004	Responsive & accessible UI	5 / DS-UX-01	WebUI	TC-UX-01	N	WCAG 2.1 AA
CSQP-NF-005	Scalability to 500 concurrent attempts	5 / DS-Perf-01	Backend / Infrastructure	TC-Scale-01	N	5,000 registered users
CSQP-SR-001	TLS 1.2+ communication	5.1.2 / DS-Sec-01	SecurityModule	TC-Sec-01	N	HTTPS required
CSQP-SR-002	Password hashing	5.1.2 / DS-Sec-01	AuthModule	TC-Sec-02	N	bcrypt/Argon2; no plaintext
CSQP-SR-003	JWT expiry & refresh rotation	5.1.2 / DS-Sec-01	AuthModule	TC-Sec-03	N	Access token ≤ 1 hour
CSQP-SR-004	API-level RBAC enforcement	5.1.2 / DS-Sec-01	AuthModule	TC-Sec-04	N	Every protected endpoint
CSQP-SR-005	Account lockout after failed logins	5.1.2 / DS-Sec-01	AuthModule	TC-Sec-05	N	15-minute lockout